Starting Methods and Starting Torque of Induction Motors

1 Basic Concept

The starting method of an induction motor depends on the type of rotor and the type of power supply.

At the instant of starting, the rotor speed is zero:

$$N = 0.$$

Therefore, the slip is

$$s = \frac{N_s - N}{N_s} = 1,$$

where N_s is the synchronous speed and N is the rotor speed.

Thus,

$$\boxed{s = 1 \quad \text{at starting}}$$

is an important relation.

2 Three-Phase Wound-Rotor Induction Motor

A wound-rotor induction motor has three-phase windings on its rotor. These rotor windings are connected to external resistors through **slip rings** and brushes.

$$\boxed{\text{Rotor winding} \rightarrow \text{Slip rings} \rightarrow \text{Brushes} \rightarrow \text{External resistance}}$$

At starting,

$$s = 1.$$

By increasing the rotor resistance R_2, the slip at which the maximum torque occurs moves toward $s = 1$.

In simplified form,

$$s_{\text{max}} \propto R_2.$$

Therefore,

$$R_2 \uparrow \quad \Rightarrow \quad s_{\text{max}} \rightarrow 1 \quad \Rightarrow \quad T_{\text{start}} \uparrow .$$

Hence, the purpose of adding external rotor resistance is to obtain a large starting torque.

$$\boxed{\text{Wound rotor} \rightarrow \text{Increase rotor resistance} \rightarrow \text{Large starting torque}}$$

The slip rings themselves do not increase the torque. Their purpose is to provide an electrical connection between the rotating rotor winding and the stationary external resistance.

3 Three-Phase Squirrel-Cage Induction Motor

In a squirrel-cage induction motor, the rotor circuit cannot be connected to an external resistance.

Therefore, starting is controlled from the **stator (power-supply) side**.

Typical starting methods are:

- Star-delta starting
- Autotransformer starting
- Inverter starting

For an induction motor, the starting torque is approximately proportional to the square of the applied voltage:

$$\boxed{T_{\text{start}} \propto V^2}.$$

Therefore, if the starting voltage is reduced,

$$V \downarrow \quad \Rightarrow \quad T_{\text{start}} \downarrow .$$

However, the starting current is also reduced.

3.1 Star-Delta Starting

The motor is connected in star (Y) during starting and changed to delta (Δ) during normal operation.

For the same line voltage V_L, the phase voltage during star connection is

$$V_{\text{ph},Y} = \frac{V_L}{\sqrt{3}}.$$

Since

$$T \propto V^2,$$

the starting torque becomes approximately

$$T_Y = \left(\frac{1}{\sqrt{3}}\right)^2 T_\Delta = \frac{1}{3} T_\Delta.$$

Thus,

$$\boxed{T_{Y,\text{start}} \approx \frac{1}{3} T_{\Delta,\text{start}}}$$

for the same line voltage.

3.2 Autotransformer Starting

An autotransformer is inserted between the power supply and the motor during starting.

It reduces the motor terminal voltage:

$$V_{\text{motor}} < V_{\text{supply}}.$$

Therefore,

$$V \downarrow \quad \Rightarrow \quad I_{\text{start}} \downarrow, \qquad T_{\text{start}} \downarrow .$$

3.3 Inverter Starting

An inverter controls both the voltage and frequency supplied to the motor.

$$\boxed{\text{Inverter} \rightarrow V \text{ control} + f \text{ control}}$$

By starting at a low frequency and appropriately controlling the voltage, the motor can start smoothly while producing useful torque.

4 Single-Phase Induction Motor

A single-phase induction motor cannot produce a starting torque using only its main winding.

At standstill,

$$\boxed{T_{\text{start}} = 0}$$

for an ideal single-phase induction motor with only the main winding.

Therefore, an auxiliary winding is used.

The current in the auxiliary winding is made to have a different **phase** from the current in the main winding.

$$I_{\text{main}} \quad \text{and} \quad I_{\text{aux}}$$

have a phase difference:

$$\phi \neq 0.$$

This phase difference produces a rotating magnetic field component, which creates starting torque.

Thus,

$$\boxed{\text{Single phase} \rightarrow \text{Auxiliary winding} \rightarrow \text{Phase difference} \rightarrow \text{Starting torque}}$$

5 Summary

Motor Type	**Starting Method**	**Starting-Torque Concept**
Three-phase wound rotor	External rotor resistance through slip rings	Increase R_2 so that the maximum-torque point approaches $s = 1$, producing a large starting torque.
Three-phase squirrel cage	Star-delta, autotransformer, or inverter	The rotor circuit cannot be externally modified. For voltage-reduction starting, $T_{\text{start}} \propto V^2$.
Single-phase induction motor	Auxiliary winding with a phase difference	Main winding alone gives zero starting torque ideally. A phase difference produces starting torque.

6 Key Relations to Remember

$$\boxed{s = 1 \qquad \text{at starting}}$$

$$\boxed{\text{Wound rotor: } R_2 \uparrow \Rightarrow s_{\max} \to 1 \Rightarrow T_{\text{start}} \uparrow}$$

$$\boxed{\text{Squirrel cage: } T_{\text{start}} \propto V^2}$$

$$\boxed{\text{Single phase: } T_{\text{start}} = 0 \quad \text{with the main winding alone}}$$

$$\boxed{\text{Auxiliary winding} + \text{phase difference} \Rightarrow \text{starting torque}}$$